

Independent Study Assignment

1) What is “Computer Architecture”?

Define the term and briefly explain how it relates to microarchitecture, ISA, and system design.

The operational framework and conceptual design of a computer system are referred to as computer architecture. It describes how hardware parts work together to carry out software commands. The architecture consists of the system design, which addresses memory hierarchy, input/output, and interconnects; the microarchitecture, which describes how these instructions are implemented; and the Instruction Set Architecture (ISA), which describes the commands a processor can comprehend. These components work together to determine programmability, power efficiency, and system performance.

2) Historical Milestones (Timeline)

Create a concise timeline (bullet list or short narrative) from early computers to modern processors. Include at least 6 major milestones.

- 1940s – The ENIAC, one of the first electronic general-purpose computers, was developed.
- 1950s – Introduction of transistor-based computers replaced vacuum tubes.
- 1960s – Development of integrated circuits revolutionized computer design.
- 1971 – Intel released the first microprocessor, the Intel 4004.
- 1980s – Rise of RISC (Reduced Instruction Set Computing) architecture improved efficiency.
- 2000s – Emergence of multi-core processors and modern GPU computing.
- 2020s – Continued innovations in AI accelerators and quantum computing research.

3) Classes of Computing Systems

Describe at least three (e.g., desktop, server, embedded). For each: typical constraints, performance goals, and example use cases.

There are several classes of computing systems, each designed for specific performance and resource requirements. Desktop computers balance performance and cost, making them ideal for general-purpose use and personal productivity. Servers prioritize reliability, throughput, and parallel processing capabilities to handle large workloads, such as database management or cloud hosting. Embedded systems, in contrast, are optimized for specific tasks within devices like cars, medical equipment, and IoT sensors, where low power consumption and real-time operation are critical.

4) Classic Components of a Computer

Explain Input, Output, Memory, Datapath, Control and how they interact during instruction execution.

Input, Output, Memory, Datapath, and Control are the traditional parts of a computer. Keyboards and sensors are examples of input devices that supply data to the system. The CPU's datapath and control unit coordinate the execution of instructions; the control unit controls data flow and sequencing, while the datapath carries out operations. Data and instructions for processing are both stored in memory. Results are displayed to people or other systems by output devices like actuators and monitors. The fetch-decode-execute cycle, which constitutes computation, is made possible by these elements working together.

5) Transistor Scaling & Moore's Law

Summarize Moore's Law and Dennard scaling at a high level and how scaling enabled performance gains.

Moore's Law, proposed by Gordon Moore in 1965, predicted that the number of transistors on a chip would double approximately every two years, leading to exponential growth in processing power. Dennard scaling complemented this trend by allowing smaller transistors to run faster without increasing power consumption. These advancements enabled faster, more energy-efficient processors for decades. However, as transistor sizes approached physical limits, the pace of scaling began to slow, leading to the search for new materials and architectures.

6) The "Power Wall"

Explain what it is, why it emerged, and its impact on clock frequency and thermal design.

The practical limits of raising CPU clock rates because of excessive power consumption and heat generation are referred to as the "Power Wall." Power density rose as transistors got smaller and more closely spaced, making efficient heat dissipation challenging. As a result, chip designers were compelled to prioritize parallel computing and energy-efficient architectures over higher clock rates. By placing a strong emphasis on multi-core designs and dynamic power management, the Power Wall significantly altered processor design objectives.

7) From Uniprocessors to Multiprocessors

Discuss why industry shifted to multi-core and the implications for software (parallelism).

After the Power Wall restricted clock growth, the necessity for continuous performance improvements drove the switch from uniprocessor to multiprocessor systems. Multiple instruction streams can run concurrently on multi-core computers, increasing efficiency and throughput. But software also had to change to accommodate this change. To efficiently use several cores, developers started designing multithreaded programs and parallel algorithms. Today, multi-core and heterogeneous computing are commonplace in data centers, game consoles, and mobile devices.

8) IC Manufacturing (Overview)

Outline the main stages from design to packaged chip (e.g., masks, lithography, deposition/etch, doping, test, packaging). A diagram is welcome (optional).

Integrated Circuit (IC) manufacturing is a complex process that transforms silicon wafers into functional chips. It begins with circuit design and the creation of photomasks. Through lithography, patterns are transferred to the wafer surface. Deposition and etching steps build up and remove material layers, while doping introduces impurities to modify electrical properties. After testing, wafers are cut into individual dies and packaged for protection and connectivity. This highly precise process requires cleanroom environments and nanometer-scale accuracy.